Manan Sharma

Software Engineer

SUMMARY

Passionate Software Engineer with a strong foundation in scalable development and a versatile command of modern programming languages. Specializes in building robust applications within Linux environments and developing end-to-end IoT solutions that process complex, real-world data. Leverages a unique background in deep learning algorithms to build intelligent, high-performance systems and integrate edge AI across diverse domains. Dedicated to writing clean, maintainable code to drive impactful technical solutions.

EXPERIENCE

Uniconverge Technologies, Noida — **IoT Engineer Intern,** 06/2026 - Present

Developed custom embedded Linux firmware utilizing OpenWrt build environments, establishing seamless interoperability across MIPS (MT7628) and ARM (NanoPi) gateway architectures. Engineered robust hardware integration for SX1303 LoRa basebands by writing custom device tree overlays, optimizing kernel-space SPI drivers, and tuning the Linux TCP/IP stack for high-throughput, low-latency edge networking.

Indian Institute of Technology, Roorkee — **Research Intern,** 06/2025 - 07/2025

Worked on Computer Vision models and data annotation tools, and developed a custom YOLOv8-based Object Detection System optimized using Intel's OpenVINO library for efficient real-time inference on NPU and accelerated performance.

EDUCATION

B.Tech Information Technology, 2024-2028

University of Jammu, Jammu & Kashmir

Higher Secondary Education, 2024

Kendriya Vidyalaya No.1, Udhampur.

TRAINING & WORKSHOPS

Junior Scientist at 26th NCSC IIT Kanpur, 2018

Participated in National Children Science Congress at Kendriya Vidyalaya Indian Institute of Technology, Kanpur.

Eye Tracking and Cognitive Science Workshop IIT Jammu, 2024

Participated in a workshop on Eye Tracking and Cognitive Science at Indian Institute of Technology Jammu, focusing on eye-tracking technologies, visual attention analysis, and cognitive behavior research.

SKILLS

Python, C, C++, Rust, R, Flutter, Kotlin, JavaScript, MATLAB, PyTorch, Keras, OpenVINO, TensorFlow, NLP, Linux/UNIX System Administration, LoRaWAN, ChirpStack, OpenWRT, Kernel Customization, Embedded Systems, Linux Firmware Development, SQL, DBMS, PowerShell, Bash, Shell Scripting, Networking, Cryptography, PCB Design, Git, Microsoft Azure, GCP, AWS, Docker, Tableau, Figma, PowerBI, Machine Learning, Deep Learning, Computer Vision, Full Stack Development, SaaS, DevOps, Raspberry Pi & Arduino Programming, Unity 3D, CAD, AR/VR Development, QGIS,

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Event Management & Organising Team Member

University of Jammu

Contributed to the successful management and execution of university events including:

- Innovathon 1.0 National level Hackathon – University of Jammu

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Licenses & Certifications

AMD (Advanced Micro Devices)

AI Agents 101: Building AI Agents with MCP and Open-Source Inference

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SPROJECTS

SlipSense: Res U-Net Based Landslide Detection and Mapping using Satellite Imagery

Developed an AI-powered landslide segmentation system using custom designed Residual U-Net (CRUNet) and DeepLabV3+ based on ASPP on 14-band multispectral satellite imagery (HDF5),exposed via a Flask REST API for inference.

YOLO Based Live Object Detection using OpenVINO

Developed a custom YOLOv8 (Using Intel's OpenVINO Library) based Object Detection System trained using COCO Dataset.

Temperature Prediction Using ETS Model: A Multi-Language Platform

This project predicts daily average temperatures using the Exponential Smoothing (ETS) model, implemented in Python, and computes discrete derivatives to analyze short-term temperature variations using C.

Custom Raspberry Pi Router | OpenWRT, Linux, Network Security

Engineered a secure, custom network router on a Raspberry Pi 3 by modifying and deploying OpenWRT Linux firmware. Hardened personal network infrastructure by configuring custom routing rules and gateway security protocols.

Implicit Detection of Food Stimuli from EEG Signals using Machine Learning

Working on an EEG-based Machine Learning system to classify implicit neural responses to food and non-food stimuli using cognitive neuroscience datasets.

Sentiment Analysis using CNN-LSTM

Working on a hybrid CNN-LSTM based Sentiment Analysis system trained on the GoEmotions dataset for multi-class emotion classification. Implemented text preprocessing, word embeddings, and deep learning architectures to capture contextual and sequential emotional patterns from textual data.